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Admittance spectroscopy of organic-silicon heterojunction structures fabricated with ultrasound assistance☆

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ABSTRACT

The development of methods for controlled modification of silicon–polymer structures is essential for advancing next-generation solar cells. This study investigates the influence of ultrasound treatment (UST) applied during fabrication on the electrical properties of Ag/PEDOT:PSS/n-Si/Zn-Ga heterostructures. The analysis combines complementary approaches based on current–voltage and capacitance–voltage measurements, as well as admittance and impedance spectroscopy. UST was found to narrow the defect energy spectrum in the interface region by restructuring unstable trap configurations and to accelerate aging processes, which are important for the long-term stability of system parameters. Quantitative characteristics of acoustically induced variations in parameters associated with both transport and relaxation processes, including characteristic times, current densities, dispersion parameters, and ideality factors, are presented. The results suggest that longitudinal vibrations primarily promote polymer phase separation, whereas radial vibrations seem to be predominantly linked to film homogenization and PEDOT molecular reorientation. It is demonstrated that ultrasound is an effective tool for controlled engineering of the properties of organic–inorganic electronic devices.

1. Introduction

Structures based on silicon and conductive polymers such as PEDOT:PSS represent one of the most promising directions in the development of next-generation hybrid heterostructure photovoltaic devices [1–3]. Their significance arises from the ability to combine the high efficiency of conventional crystalline silicon with the low cost and simple processing of organic materials [4,5]. In particular, Si/PEDOT:PSS structures can be fabricated at temperatures below 200 °C using relatively simple deposition techniques, including spin coating and spray coating, which substantially reduces energy consumption and overall production costs [1,2,5–7]. Owing to its high work function, PEDOT:PSS forms a *p–n* junction when deposited onto *n*-Si, thereby enabling efficient separation of photogenerated charge carriers without the need for complex high-temperature diffusion or ion implantation processes [1,2,5,8]. This approach simplifies the device architecture and reduces the optical losses typically associated with heavily doped layers [3,9]. Furthermore, the compatibility of PEDOT:PSS with roll-to-roll technologies makes it highly attractive for the large-scale production of organic electronic devices [3,5].

The properties of hybrid Si/PEDOT:PSS structures are governed by a complex interplay of interrelated factors. Among the most important are the surface state density, which determines the formation of the inversion layer [7]; the electrical conductivity of the polymer, governed by the internal structure and morphology of the film [1,2]; and the contact uniformity, since the formation of interfacial voids leads to enhanced recombination [5,10]. Controlled modification of these parameters can be achieved using a variety of approaches, including chemical modification of the polymer [1,2], interface engineering through the introduction of intermediate layers [3,11], optimization of annealing conditions [12], and post-treatment procedures involving light soaking or electrical biasing [13].

Ultrasound (US) represents another effective tool for tuning the properties of heterostructures. More broadly, it is widely employed in the fabrication of electronic devices. US can be used either as a standalone technique for specific applications, such as the rapid preparation of homogeneous precursor solutions for metal oxide semiconductors and dielectrics [14], the sonochemical synthesis of photoactive materials [15], or contact formation [16–18], or as an auxiliary element integrated into conventional technological processes. Examples of the latter include ion implantation, where US promotes mechanical stress relaxation and reduces defect density [19,20], and chemical etching during porous layer formation, where it improves structural uniformity while reducing hydrogenation and oxidation levels [21].

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In the field of organic electronics, ultrasound treatment (UST) of solutions is widely employed for the synthesis of polymer nanocomposites [22–24]. During the fabrication of polymer films, including PEDOT:PSS, ultrasonic spray coating represents one of the most common applications of acoustic waves [25–27]. An alternative approach involves the excitation of elastic vibrations directly within the substrate onto which the film is deposited. In this context, two configurations are possible: the application of acoustic vibrations during spray deposition [22,28–30], or post-deposition UST while the film remains wet after deposition by methods such as spin coating [29–37]. Such treatment increases the density and uniformity of the resulting film, reduces porosity, and enhances electrical conductivity.

At the same time, it should be noted that the vast majority of studies focus primarily on the properties of films fabricated using UST rather than on the characteristics of devices incorporating them. To the best of our knowledge, only a limited number of reports have examined fully assembled systems, such as solar cells [28,37,38]. However, even in these studies, the primary focus was placed on integral electrical characteristics, such as open-circuit voltage, short-circuit current, and fill factor, whereas the influence of UST on the microscopic charge transport mechanisms was only rarely analyzed [38]. The aim of this work was to investigate the electrical properties of Si/PEDOT:PSS structures subjected to UST during polymer layer formation in the post-treatment regime. For this purpose, both DC and AC measurements were employed, enabling the deep analysis of characteristics associated with charge transport and recombination processes. Furthermore, in this study, ultrasonic vibrations were utilized at frequencies that exceeded by 1–2 orders of magnitude those (13–40 kHz) used in previous investigations.

2. Experimental details

In the present work, Ag/PEDOT:PSS/n-Si/Zn-Ga heterojunction structures were fabricated, and the corresponding manufacturing flow is presented in Fig. 1. Double-side-polished *n*-type Si (100) phosphorus-doped wafers with a resistivity of 1–10 Ω cm and a thickness of 525 μm were used as substrates. The Si samples (1×1 cm^2) were immersed in 30% hydrofluoric acid (HF) to remove the native silicon oxide layer. An aqueous solution of poly(3,4-ethylenedioxythiophene):polystyrene sulfonate (PEDOT:PSS) supplied by Sigma-Aldrich (655201, Germany) was used for the formation of the *p*-type layer. The polymer thin film was deposited by spin coating for 30 s using two different rotation speeds: 3000 and 5000 rpm. It is well established that the rotation speed influences the efficiency of solvent removal, the film microstructure, and, most importantly, the final film thickness. For example, Ref. [37] reported PEDOT:PSS film thicknesses of approximately 52 and 40 nm for films fabricated at 3000 and 5000 rpm, respectively.

Subsequently, the structures underwent room-temperature relaxation holding for 20 min. The primary objectives of this stage were film leveling after spin-coating due to solution redistribution and the smoothing of thickness inhomogeneities; gentle removal of the residual solvent, where slow water desorption prevents the formation of microcracks and porosity that can occur during abrupt heating; partial phase segregation of the PEDOT and PSS phases along with the rearrangement of conductive domains; and the reduction of internal mechanical stresses, which increases film stability during subsequent annealing. UST was applied to a subset of samples specifically during this stage. For its implementation, the backside of the silicon substrate with the wet polymer layer was placed on a cylindrical piezoelectric transducer (PZT ceramic, 0.958 mm in thickness and 19.9 mm in diameter). Acoustic coupling between the sample and the transducer was provided by a dedicated acoustic gel. An alternating voltage with an amplitude of approximately 10 V was applied across the transducer to excite the oscillations. The excitation frequency was selected to enable UST in two different oscillation modes. In the first case, the frequency was matched to the thickness resonance of the transducer (2.5 MHz), whereas in the second case, it corresponded to the radial resonance (503 kHz, third harmonic). The thickness-resonance excitation generated primary waves dominated by displacement normal to the transducer surface (uP mode), whereas the radial resonance produced predominantly radial oscillations with significant shear components and a more complex spatial distribution of displacement (uR mode). The acoustic intensity within the silicon substrate was approximately 0.6 W/cm^2 for the uP mode and 0.4 W/cm^2 for the uR mode, with corresponding atomic displacement amplitudes of approximately 2 and 4 nm, respectively. These comparable acoustic excitation conditions provide a basis for attributing the observed differences in the effects of UST, discussed below, primarily to the specific vibration mode. UST was maintained continuously throughout the 20-min room-temperature relaxation phase. During treatment, the transducer-sample assembly was mounted on a heat sink, ensuring that the temperature increase of the investigated structures did not exceed 7 K.

The subsequent step involved thermal annealing at 140 °C for 15 min under ambient atmospheric conditions, ensuring efficient removal of the residual solvent and stabilization of the structural properties. To evaluate the electrical characteristics, a silver contact was thermally evaporated onto the PEDOT:PSS surface, while a Zn/Ga eutectic was applied to the backside of the silicon substrate to form the back contact.

Consequently, several series of samples were prepared, differing in polymer layer thickness, the presence or absence of UST during the relaxation stage, and the type of acoustic oscillations applied. To ensure reproducibility and assess the statistical significance of the experimental results, each experimental series comprised 3–7 independently fabricated and tested devices. Outliers were identified and excluded using the standard Grubbs' test at a significance level of 0.05; such outliers may be associated with fabrication defects, such as edge short circuits or micro-cracks. The electrophysical parameters reported below are mean values calculated across all devices in each series, with error bars indicating one standard deviation. The series nomenclature included the spin-coating speed (in thousands of rpm), the relaxation time (in minutes), and the letter designation of the UST mode. The labels for each series are shown in Fig. 1, whose inset also provides a schematic illustration of the device architecture.

The structures were characterized by capacitance–voltage (C–V), capacitance–frequency (C–f), and impedance spectroscopy (IS) measurements performed in the dark using a Sourcetronic ST2829C LCR meter with a 10 mV AC signal applied to the device. The C–f and IS measurements were carried out over a frequency f range from 200 Hz to 1 MHz, whereas the C–V measurements were performed at frequencies of 1 MHz, 100 kHz, and 10 kHz. The C–V and IS measurements were conducted at bias voltages V_b ranging from −1.5 V to 1.5 V and from −1.2 V to 0 V, respectively, while the C–f measurements were performed were taken at V_b values of 0 and 0.4 V. The C–V and IS measurements were carried out at room temperature, whereas a subset of the C–f measurements was performed in the 200–300 K temperature range. The sample temperature was monitored using a PT100 sensor. Additionally, current density–voltage (J–V) characteristics were recorded in the dark at $T = 296$ K using a Keithley 2450 source measure unit.

3. Results and discussion

3.1. C-V measurements

Fig. 2a presents typical experimental capacitance–voltage characteristics of the Ag/PEDOT:PSS/n-Si/Zn–Ga diode measured at modulation frequencies of 1 MHz, 100 kHz, and 10 kHz at $T = 296$ K. As can be seen, a pronounced maximum appears on the forward-bias branch, which is associated with the onset of efficient minority carrier (hole) injection, leading to enhanced recombination and a subsequent sharp decrease in capacitance [39–41]. In general, the peak voltage V_p

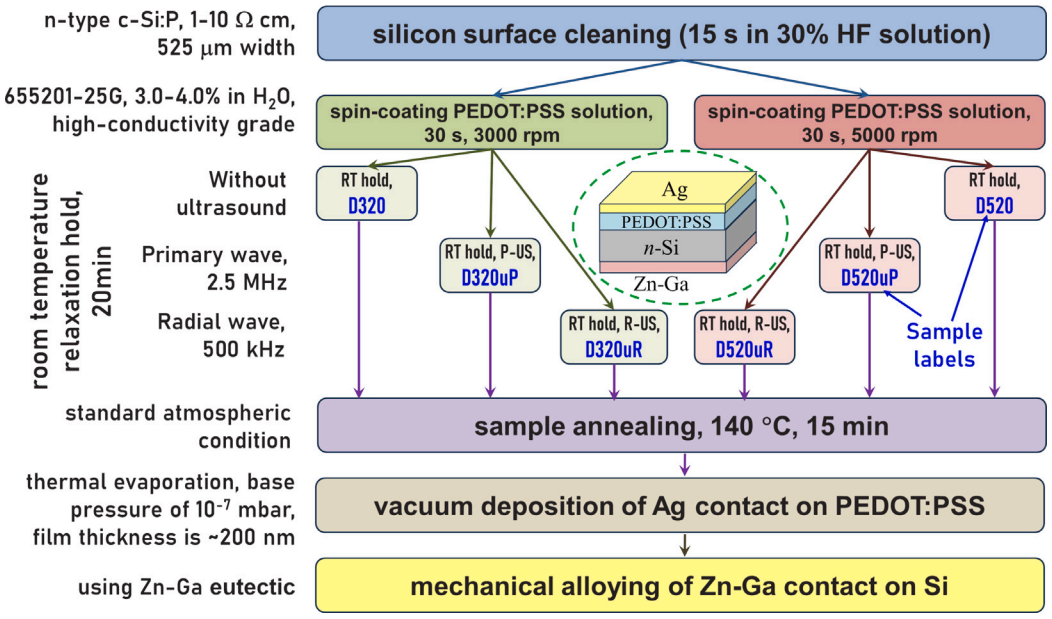


Fig. 1. Fabrication procedure for Ag/PEDOT:PSS/n-Si/Zn-Ga structures. Inset: sketch of the device structure.

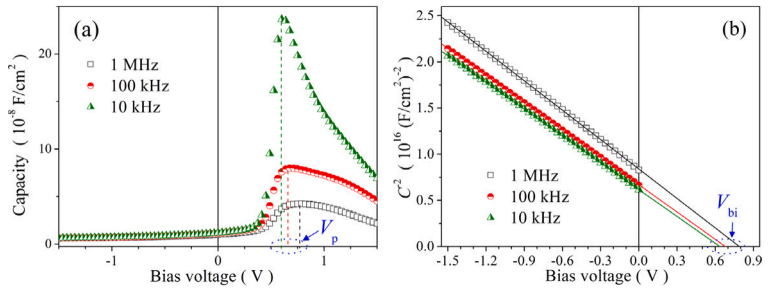


Fig. 2. (a) C–V characteristics of the D320 sample measured at different frequencies. (b) Mott–Schottky plots for the D320uR sample. Symbols represent the experimental data, while the lines correspond to linear fits.

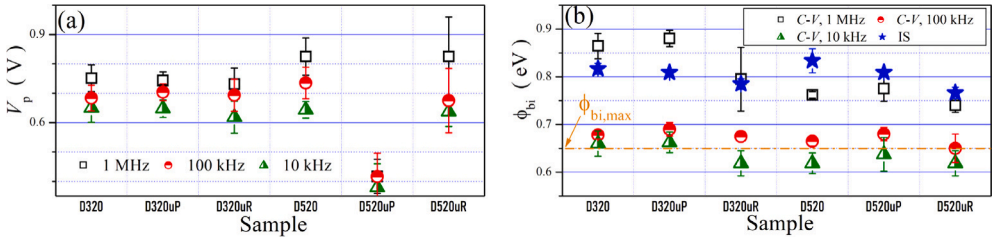


Fig. 3. Values of the peak voltage (a) and built-in potential (b) for different sample series. The presented built-in potential values were determined from both *C–V* measurements performed at different frequencies and impedance spectroscopy measurements (indicated by stars). The $\phi_{bi,max}$ value is indicated by the dash-dotted line.

corresponding to this maximum increases with increasing measurement frequency [39], in agreement with the experimental results shown in Fig. 2a.

The V_p values obtained for the different sample series are summarized in Fig. 3a. It is evident that, for thicker layers, UST does not significantly affect the peak voltage. In contrast, for thinner polymer layers, longitudinal vibrations lead to a substantial decrease in V_p. This behavior may be associated with acoustically induced (AI) phase separation of PEDOT and PSS near the interface [32], potentially facilitating the formation of intimate local contacts between conductive PEDOT nanocrystals and silicon atoms. In thicker layers, the gel matrix may damp local microdisplacements at the interface, thereby limiting such rearrangements. Conversely, radial vibrations may produce a less pronounced alternating normal pressure at the interface, which could be insufficient to overcome the Coulomb barrier between PEDOT and PSS.

The reverse-bias branch was used to estimate the built-in potential ϕ_{bi} using the Mott–Schottky method (see Fig. 2b). Specifically, $\phi_{bi} = qV_{bi}$, where V_{bi} was determined from the intercept of the linear fit to the C^{-2} versus bias voltage dependence on the voltage axis. Fig. 3b summarizes the obtained ϕ_{bi} values. Prior to discussing the effect of UST on the built-in potential, it should be noted that the maximum value of this parameter is determined by the work-function difference between PEDOT:PSS and silicon:

$$\phi_{bi,max} = \Phi_{PEDOT:PSS} - \Phi_{Si} = \Phi_{PEDOT:PSS} - \chi_{Si} - kT \ln\left(\frac{N_C}{N_d}\right), \quad (1)$$

where $\chi_{Si} = 4.05$ eV is the electron affinity of silicon, and the last term describes the position of the Fermi level relative to the conduction-band minimum in silicon. Here, N_C is the effective density of states in the conduction band, and N_d is the donor concentration. The N_d was estimated from Mott–Schottky plot and the obtained value, $N_d = (1 - 2) \times 10^{14}$ cm^{-3}, was consistent with the nominal conductivity specifications of the used substrates. Thus, $\Phi_{Si} = (4.36 \pm 0.01)$ eV. In turn $\Phi_{PEDOT:PSS}$ is known [42] to depend on the PEDOT:PSS ratio, which in the present case (conductivity > 200 S/cm) was approximately 1:2.5. Consequently, $\Phi_{PEDOT:PSS} = 5.1$ eV [42,43], yielding $\phi_{bi,max} = (0.65 \pm 0.01)$ eV. As can be seen from the data presented in Fig. 3b, the obtained values exceed this limit often. Therefore measured C is determined not only by the space-charge capacitance but also contains an additional contribution most likely associated with interface-state capacitance and/or the presence of an insulating interlayer (the so-called Helmholtz capacitance) [44]. The influence of this additional capacitance increases with increasing measurement frequency; therefore, while the values obtained at 10 kHz can be used to estimate the built-in potential, the results at 1 MHz can be used to evaluate additional capacitance.

At low frequencies, the obtained ϕ_{bi} values are nearly independent of UST and closely approach the theoretical limit. This observation indicates that acoustic oscillations do not modify either the work function of PEDOT:PSS or the initial energy offset, which is fully consistent with expectations. In contrast, the interface-state capacitance derived from the 1 MHz data decreases noticeably after radial UST compared with the control sample, with a more pronounced effect observed for thicker films. These results may suggest that radial waves propagating along the interface do not strongly promote PSS redistribution into the bulk of the layer, but may instead suppress structural inhomogeneity at the junction interface. However, in the absence of significant vertical PEDOT/PSS phase separation, the hole-injection barrier may remain largely unchanged.

3.2. Admittance spectroscopy

It is well established [45,46] that C–f measurements enable estimation of the trap density of states (tDOS) in barrier structures. Within the formalism proposed by Walter et al. the tDOS profile can be determined using the following differential relationship [45,47,48]:

$$n_t(E_f) = -\frac{V_{bi}}{qWkT}\frac{f\,dC}{df}, \tag{2}$$

where W is the depletion width

$$W = \sqrt{\frac{2\varepsilon\varepsilon_0(V_{bi} - V)}{qN_d}}, \tag{3}$$

and demarcation energy E_f is defined by

$$E_f = kT\ln\left(\frac{f_0}{f}\right) = kT\ln\left(\frac{\nu_{00}T^2}{f}\right), \tag{4}$$

where f_0 is the attempt-to-escape frequency, which depends on the density of states in the conductive band, the carrier thermal velocity, and the trap capture cross-section [49], ν_{00} is the reduced attempt frequency, which depends on the material properties and the specific defect type and therefore generally requires independent determination. It is well established [49–51] that, at the frequency f_{max} corresponding to the maximum of the $-fdC/df$ dependence, the following relationship holds:

$$f_{max} = \nu_{00}T^2\exp(-E_f/kT). \tag{5}$$

Accordingly, analysis of the temperature dependence of f_{max} enables estimation of the ν_{00} value.

Fig. 4a presents the measured temperature-dependent C–f characteristics, while Fig. 4b shows the corresponding $-fdC/df$ curves illustrating the shift of the peak position. Fig. 4c displays the temperature dependence of $f_{max}T^{-2}$, which exhibits the expected linear behavior on a semilogarithmic scale in accordance with Eq. (5). Linear fitting of the dependence shown in Fig. 4c yielded $\nu_{00} = (4.3 \pm 0.2) \cdot 10^4$ s^{-1}K^{-2}. Notably, the obtained ν_{00} value corresponds to an attempt-to-escape frequency of $3.9 \cdot 10^9$ Hz, in good agreement with values reported previously for organic materials [49,51,52].

Fig. 5 presents the energy profiles of the tDOS calculated according to Eqs. (2)–(4) from room-temperature C–f measurements at zero bias for three sample series. The calculations used V_{bi} values extracted from C–V measurements at 10 kHz. The resulting dependencies are representative of the other sample series and bias conditions.

In organic semiconductors, which typically exhibit lower crystallinity than inorganic materials, the assumption of discrete trap levels is often overly simplistic [53–55]. Consequently, Gaussian distributions are frequently employed to model the experimentally determined tDOS, particularly for data obtained from C–f measurements [46,56–60]. However, as illustrated in Fig. 5, the obtained profiles cannot be adequately described by a single Gaussian function. This suggests the presence of multiple trap types associated with defects of different structural origins. A well-established and widely used strategy for decomposing experimentally determined tDOS spectra into distinct trap-state contributions consists of using a superposition of multiple Gaussian components [53,61–63]:

$$n_t(E_f) = \sum_{i=1}^{N}\frac{N_{t,i}}{\sqrt{2\pi\sigma_{t,i}^2}}\exp\left[-\frac{(E_f - E_{0,i})^2}{2\sigma_{t,i}^2}\right], \tag{6}$$

Eq. (6) accounts for the presence of N distinct trap types, where $N_{t,i}$, $E_{0,i}$, and $\sigma_{t,i}$ denote the trap concentration, mean energy, and energy distribution width, respectively, for the ith defect type. Eq. (6) was used to fit the experimental data. The parameters of the individual Gaussian components were determined by minimizing the squared error fitness function. The optimization was performed using the simplified teaching–learning-based optimization algorithm [64], a metaheuristic optimization method that has demonstrated good performance in multi-parameter optimization problems [65]. The number of Gaussian components, N, was selected to balance two competing criteria: (i) maximizing the agreement between the fitting function and the experimental curve, and (ii) minimizing the number of components used. The complete methodological details are provided in the Supplementary Material. For the selected number of Gaussian components, the mean discrepancy between the experimental and fitted curves remained approximately 2% across all samples. The uncertainties in determining the component parameters, namely the peak positions and relative contributions, represented by the error bars in Figs. 6 and 7, were primarily determined by the sample-to-sample variation within each series. In contrast, the fitting uncertainty for an individual curve was substantially lower, reaching approximately 2 meV for the peak positions and 0.2% for the relative contributions.

This study focused on determining the number of defect types as a function of the fabrication procedure, as well as the positions of the maxima of the corresponding distributions and the relative contribution of each defect type to the total signal. In the latter case, the ratio of the area under each individual Gaussian component, S_i, to the total area under the $n_t(E_f)$ curve was used as a quantitative measure of the contribution δ_i:

$$\delta_i = S_i / \sum_{i=1}^{N} S_i, \tag{7}$$

where

$$S_i = \int_{E_{min}}^{E_{max}}\frac{N_{t,i}}{\sqrt{2\pi\sigma_{t,i}^2}}\exp\left[-\frac{(E_f - E_{0,i})^2}{2\sigma_{t,i}^2}\right]dE_f = \frac{N_{t,i}}{2}\cdot\left[\mathrm{erf}\left(\frac{E_{max} - E_{0,i}}{\sqrt{2}\sigma_{t,i}}\right) - \mathrm{erf}\left(\frac{E_{min} - E_{0,i}}{\sqrt{2}\sigma_{t,i}}\right)\right]. \tag{8}$$

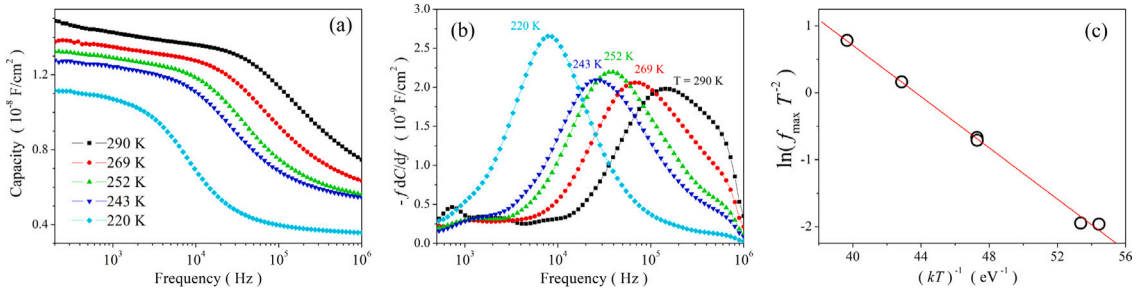


Fig. 4. Frequency dependences of (a) capacitance and (b) the $-f \cdot dC/df$ derivative measured at different temperatures. (c) Arrhenius plot of the $-f \cdot dC/df$ peak frequencies versus $1/kT$. Symbols represent the experimental data, while the lines correspond to linear fits. Sample of D520 series.

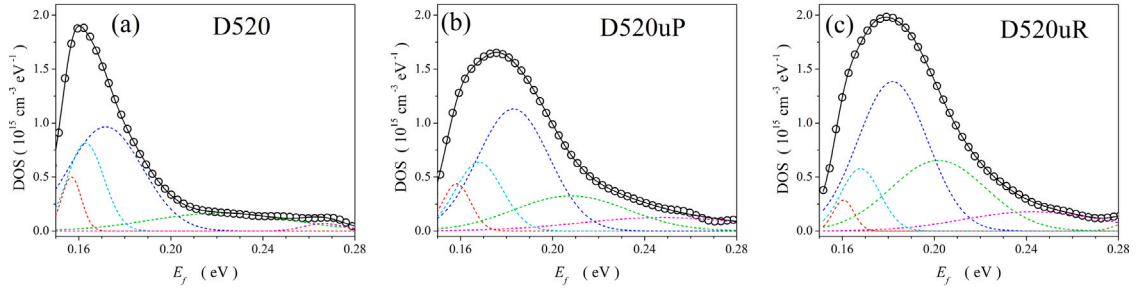


Fig. 5. tDOS profiles for samples D520 (a), D520uP (b), and D520uR (c) obtained at zero bias. Symbols represent the experimental data, dashed lines denote Gaussian fits associated with individual defect types, and solid lines correspond to the overall fit obtained using Eq. (6).

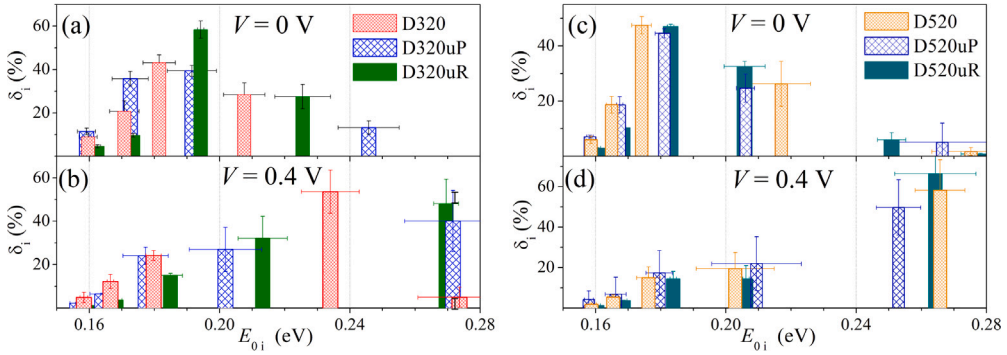


Fig. 6. Relative contribution spectra of trap components in structures with thick (a, b) and thin (c, d) polymer layers, measured at zero bias (a, c) and under a 0.4 V bias (b, d). Filled rectangles correspond to structures fabricated under radial acoustic oscillations, sparse cross-hatching denotes longitudinal oscillations, and dense cross-hatching represents control samples prepared without ultrasound treatment.

E_{min} and E_{max} are the energy limits of the analyzed defect distribution, determined by the frequency range of the capacitance measurements, and erf denotes the error function. The Romberg method was used to evaluate erf.

Fig. 6 summarizes the results obtained for all sample series under different bias conditions. Under forward bias, the width of the space charge region (SCR) decreases, shifting the probing region closer to the *n*-Si/PEDOT:PSS interface. Consequently, at a forward bias of 0.4 V, the measurements become more sensitive to states located directly at the interface or within a thin near-interface region.

Two primary groups of traps are consistently observed across all samples: shallow (E_f ~ 0.16–0.20 eV) and deep (E_f ~ 0.22–0.28 eV) states. Under forward bias, the contribution of the deep states increases markedly in all series, confirming their localization at the Si/PEDOT:PSS interface. Regardless of the layer thickness, radial UST significantly suppresses the contribution of shallow defects (0.16–0.17 eV) in favor of a more localized level at approximately 0.19 eV. In the vicinity of the interface, an increased contribution of the deepest level is observed in the uR-UST mode. Specifically, in the D520uR sample series, this contribution reaches a maximum of 66% among all investigated specimens. This behavior supports an acousto-induced smoothing effect that reduces energetic disorder of the traps and leads to a more homogeneous interface. Furthermore, these findings are consistent with previously reported effects of improved spreading, wettability, and suppression of macroscopic inhomogeneities in polymer films under an acoustic field [29,32,33].

When longitudinal waves are used, this effect is less pronounced, consistent with previously reported [29] higher efficacy of horizontal acoustic vibrations in improving film homogeneity. Nevertheless, both longitudinal and radial waves exhibit similar trends, characterized by: (i) a substantial increase in the contribution of deep traps (~ 0.28 eV) near the interface in thick films, arising from the reconfiguration of defects with an activation energy of 0.23 eV; and (ii) a reduction in the trap activation energy within the SCR region from 0.21–0.22 eV by several tens of meV.

Energy levels in the 0.16–0.19 eV range have been reported in silicon in association with oxygen-related vacancy complexes (A-centers) [66,67], among other possible defect configurations. The observed

evolution of this spectral range under acoustic treatment may therefore be consistent with changes in the occupation and/or local environment of such vacancy-related defects, potentially promoted by localized heating and microstresses. The resulting reduction in the energetic dispersion and the emergence of a more distinct level at approximately 0.19 eV may thus reflect AI modification or redistribution of defect configurations. However, this assignment should be regarded as a plausible interpretation rather than a definitive identification, since unambiguous determination of the microscopic origin of these states requires complementary structural characterization.

In our opinion, the notable similarity between the aging-induced evolution of the tDOS in organic structures and the effects of ultrasound treatment warrants attention. Specifically, Liu et al. [56] reported that tDOS peaks located near 0.22–0.23 eV in fresh PEDOT devices shift toward 0.27–0.28 eV upon prolonged aging (~ 1000 h, under hermetic encapsulation in the dark at room temperature). Remarkably, a comparable shift is observed in our samples after only 20 min of ultrasonic treatment at room temperature (Fig. 6b). This similarity suggests that UST may significantly accelerate the system's evolution toward its thermodynamically stable state.

Furthermore, in our view, the levels with the highest activation energies (0.27–0.28 eV) can reasonably be attributed to dangling-bond defects, such as P_b centers. Although such defects are traditionally associated with the Si/SiO_x interface and the oxide layer was removed by hydrofluoric acid etching prior to polymer deposition in our work, it is well established that a thin SiO_x layer can readily reform after PEDOT:PSS deposition on Si, thereby constituting an important factor limiting the quality of interface passivation [68,69]. The presence of such defect species in Si/PEDOT:PSS structures has also been reported previously [7] and is therefore plausible in the present structures. This interpretation is further supported by the close agreement between the extracted activation energies and the characteristic energy levels reported for P_b centers [70,71], as well as by the interfacial localization of these traps (see Fig. 6). Moreover, since passivation in Si/PEDOT:PSS systems is largely governed by PSS and is accompanied by surface oxidation of Si [72], the deep traps identified in this work may be more closely associated with P_{b1} defects, or structurally similar oxygen-containing configurations [73,74], whose reported energy levels provide a closer match to our experimental values [70]. The increased contribution of these levels following UST may, in turn, be related to acoustically enhanced oxygen incorporation at dangling-bond sites, potentially facilitated by silicon lattice vibrations and enhanced oxygen diffusion under acoustic excitation.

The total trap concentrations per unit volume were also estimated as follows:

$$N_{total} = \sum_{i=1}^{N} N_{t,i} \,. \tag{9}$$

The results are presented in Fig. 7. As shown, the defect concentration in the silicon–polymer contact region is higher than that in the near-surface silicon layer, as expected. During annealing and relaxation, a larger volume of the wet gel may undergo significant shrinkage, potentially inducing stronger localized microstresses at the substrate interface and increasing the concentration of structural defects and traps within the SCR. Consequently, the overall trap density in the SCR is lower for thinner layers. Furthermore, UST has negligible effect on the total number of these traps, indicating that the differences observed in Figs. 5a and 5c are primarily driven by AI defect reconfiguration.

Directly at the interface of thick layers, the application of UST (in both uP and uR modes) leads to a slight increase in the total interface trap density. This is consistent with considered model in which liberated insulating PSS macromolecules become trapped near the interface due to the high viscosity of the thick layer, which marginally increases parasitic capacitance and impedes charge injection. In contrast, for thinner layers, radial UST slightly reduces the overall defect concentration; however, as shown in Fig. 5d, this reduction is primarily driven

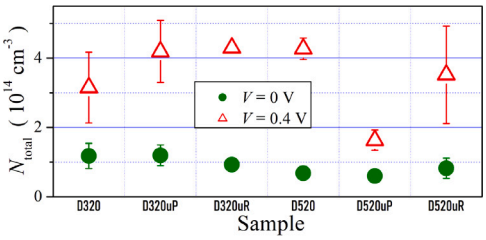


Fig. 7. Total trap concentration for different sample series.

by acoustic annealing of defects within the near-surface silicon layer. Concurrently, longitudinal waves substantially reduce the total trap density, effectively cleaning the interface, in agreement with previous reports [34]. Specifically, perpendicular alternating stresses in the thin, wet polymer layer may disrupt PEDOT Coulombic interactions, potentially promoting phase separation and redistribution of the insulating PSS phase away from the silicon surface into the bulk of the film [29,32]. As a result, an ultrathin, structurally well-defined layer of highly conductive PEDOT nanocrystallites could form at the interface, facilitating hole injection.

3.3. Impedance spectroscopy

Typical Nyquist plots showing the imaginary versus real components of the complex impedance under different bias voltages for the investigated structures are presented in Figs. 8a and 8b. Generally, numerous models can be used to describe the impedance of barrier structures [40,75–78], and the appropriate model should be selected based on its ability to accurately reproduce the system's AC response. As shown in Fig. 8b, deviations from the semicircular behavior characteristic of a single dominant process emerge in the high-frequency region. Accordingly, the equivalent circuit shown in Fig. 8c was selected to model the experimental data. This model not only accurately reproduces the characteristic features of the Nyquist plots, as indicated by the solid lines in Figs. 8a and 8b, but is also widely used for organic-based structures [38,46,56,76,79–82].

The $R_{s,IS}$ element represents the resistive losses of the entire device, including contact resistance [80]. The R_t element, coupled with CPE_t, is associated with bulk charge-transport resistance and governs the high-frequency response of the Nyquist plot [56,80,82]. The $R_r \parallel CPE_r$ network corresponds to the recombination resistance and chemical capacitance associated with interfacial charge-transfer processes [56,76,79]. These processes exhibit longer relaxation times and therefore dominate the low-frequency response.

The equivalent circuit incorporates constant phase elements (CPEs) instead of ideal capacitors to account for structural and energetic inhomogeneities, such as interface roughness and the energy distribution of trap states [40,83]. The CPE is a generalization of the ideal capacitive element and is used to describe non-ideal electrical processes occurring in the device [40,75,78,79]. The impedance of a CPE is defined as $Z_{CPE} = Q^{-1}(i\omega)^{-p}$, where p is the dispersion parameter, whose deviation from unity characterizes the degree of system inhomogeneity. Fig. 9a presents the dispersion parameters associated with the different processes for the investigated sample series. Notably, both p_t and p_r showed no noticeable bias dependence. As shown, $p_t \simeq 1$. This indicates that bulk carrier transport within PEDOT:PSS is highly homogeneous and that the polymer matrix behaves as an ideal bulk capacitance, remaining virtually unaffected by ultrasound. The parameter p_r reflects the degree of spatial and energetic uniformity at the interface. Following longitudinal UST, a clear increase in this parameter is observed, particularly for the thin film. As discussed above, alternating pressure directed perpendicular to the interface may promote phase separation and formation of a more homogeneous contact. In contrast, radial UST

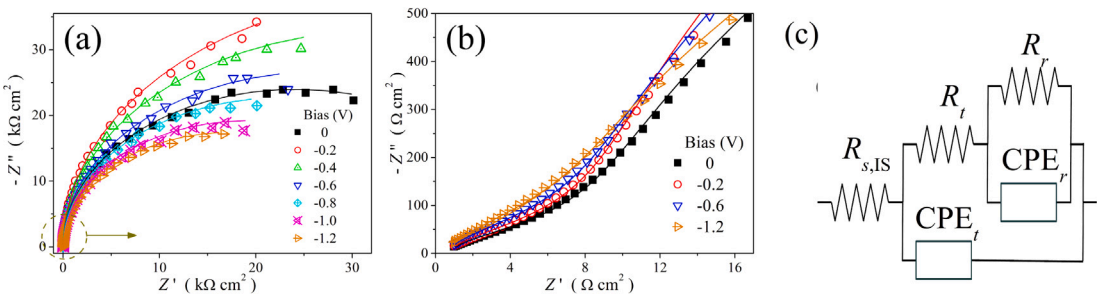


Fig. 8. (a) Nyquist plot of D320uR sample for various bias. Symbols represent the experimental data, lines correspond to the fit obtained using equivalent circuit model. (b) High frequency portion of 8(a). (c) Equivalent circuit model used to approximate the IS measurements.

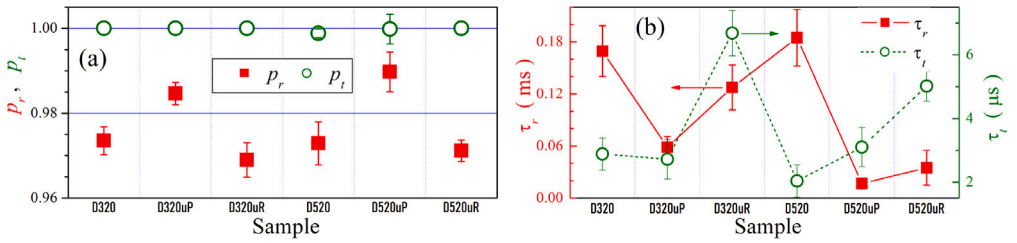


Fig. 9. Dispersion parameters (a) and characteristic times (b, $V = -0.2$ V) for different sample series. Open circles and filled squares correspond to the transport and recombination processes, respectively.

may improve the overall macroscopic uniformity of the film without inducing significant vertical PEDOT/PSS phase separation. Consequently, a disordered mixture of conductive and insulating domains may remain at the silicon interface, potentially preserving a high degree of local potential inhomogeneity and thereby limiting improvement of the p_r. Notably, an increase in p_r induced by longitudinal acoustic vibrations has also been reported previously [38].

Knowing R, Q, and p, the characteristic time of the corresponding process can be estimated: $\tau = (RQ)^{1/p}$. It was found that both τ_r and τ_t depend on the applied bias voltage, and the values obtained at $V = -0.2$ V are presented in Fig. 9b. As shown, the characteristic time of bulk carrier transport, τ_t, remains nearly unchanged within the experimental uncertainty upon application of longitudinal acoustic waves. In contrast, the horizontal displacement component associated with radial UST may promote partial orientation of PEDOT macromolecules parallel to the interface. Such an orientation could induce conductivity anisotropy, consistent with the approximately 2.4-fold reduction in the vertical charge-transport time observed irrespective of film thickness. The influence of UST on the recombination time, τ_r, is considerably more pronounced. A decrease in τ_r of up to one order of magnitude is observed, with the effect becoming stronger as the layer thickness decreases and upon transition from radial to longitudinal vibrations. This behavior may be attributed to acoustically induced artificial aging associated with the emergence of faster and deeper recombination centers, as revealed by the admittance spectroscopy analysis. The effect is further enhanced by reduced damping within the gel-like polymer matrix and facilitated minority-carrier injection.

The effective capacitance associated with transport processes was also calculated as $C_t = Q_t^{\frac{1}{p_t}} R_t^{\frac{1-p_t}{p_t}}$, and the built-in potential was determined from Mott–Schottky plot (see Fig. 4b). The ϕ_{bi} values obtained from IS measurements are in close agreement with those derived from C–V characteristics at 1 MHz, both in absolute magnitude and in the minor influence of UST in the uR mode. This agreement arises from the fact that the $R_t \parallel \mathrm{CPE}_t$ combination accurately describes the high-frequency response of the bulk and the region adjacent to the interface.

Fig. 10 presents the results of the series resistance determination. The series resistance $R_{s,IS}$ obtained from impedance spectroscopy is

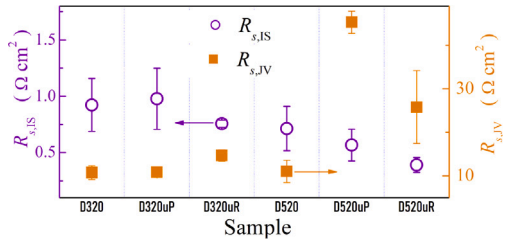


Fig. 10. Series resistance values obtained from IS (open circles) and J–V measurements (filled squares) for different sample.

defined as the intercept of the Nyquist plot in the limit $\omega \to \infty$. In such high-frequency regime, all capacitive elements of the structure are effectively shunted, and charge carriers are not required to overcome barriers between conductive PEDOT grains, allowing intragrain transport to dominate. Consequently, the measurement reflects purely ohmic losses in the conductive components, including metal contacts, probes, interconnects, and the silicon substrate. Given that the resistance of the silicon wafer is below 0.3 Ω cm^2, the measured values of 0.7–0.9 Ω cm^2 are primarily attributed to contact resistance. For thick films, radial ultrasound slightly reduces series resistance, consistent with improved wetting and enhanced adhesion of the liquid phase to the substrate [32,33], effects that are particularly relevant at high frequencies. For thin films, the initial (control) resistance is lower due to better compaction at higher spin-coating speeds, and UST further reduces this value. Overall, the results indicate that radial acoustic waves are most effective in increasing the real contact area at the metal/silicon interface and its coupling to the conductive polymer domains.

3.4. Current–voltage characteristics

Fig. 11 presents representative J–V characteristics of the investigated structures. One can see that the application of UST leads to a modification of the J–V curve shape. To analyze the results, a double-diode model was employed (see the inset in Fig. 11), according to

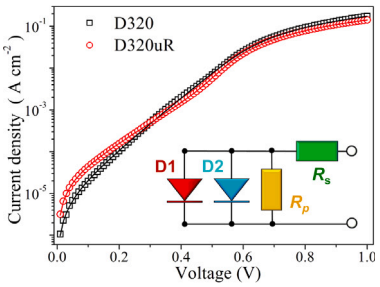


Fig. 11. Forward bias current–voltage characteristics for samples D320 (squares) and D320uR (circles). Symbols represent the experimental data, lines correspond to the fit obtained using Eq. (10). Inset: equivalent circuit model used to approximate the J–V characteristics.

which [7,84]:

$$J = J_{01}\left\{\exp\left[\frac{q\left(V - JR_{s,\mathrm{JV}}\right)}{kT}\right] - 1\right\} + J_{02}\left\{\exp\left[\frac{q\left(V - JR_{s,\mathrm{JV}}\right)}{nkT}\right] - 1\right\} + \frac{V - JR_{s,\mathrm{JV}}}{R_p}. \quad (10)$$

where the first term is associated with the diffusion-limited transport of carriers injected across the barrier, the second describes the overall SCR recombination, J_{01} and J_{02} are the saturation current densities, n is the ideality factor, and $R_{s,\mathrm{JV}}$ and R_p are the series and shunt resistances, respectively.

It should be noted that $R_{s,\mathrm{JV}}$ is determined under DC conditions, where carriers must traverse the entire structure, overcoming, in particular, the potential barriers between the conductive PEDOT grains. Consequently, this value also accounts for the integral resistance of the entire PEDOT:PSS polymer layer; hence, $R_{s,\mathrm{JV}} > R_{s,\mathrm{IS}}$ (see Fig. 10). The AI changes in $R_{s,\mathrm{JV}}$ correlate with the effects observed for τ_t (Fig. 9b), namely, the hindered current flow after radial UST due to the reorientation of PEDOT molecules. The significant increase in $R_{s,\mathrm{JV}}$ detected for the D520uP sample is, in our opinion, associated with the efficient separation of polymer phases specifically within the thin sample. Directly at the interface, this facilitates hole injection (Fig. 3a) and reduces the trap density (Fig. 7); however, a highly resistive, local PSS-enriched sublayer is formed within the polymer bulk, leading to a corresponding increase in $R_{s,\mathrm{JV}}$. At high frequencies, this sublayer is simply shunted, which explains why the increase in τ_t is substantially less pronounced.

The behavior of J_{01} and R_p across different sample series exhibits several common features (see Fig. 12a). The diffusion current density component is lowest in control samples fabricated without ultrasonic treatment. In addition, for the D320 series, J_{01} is several orders of magnitude lower than in samples with thinner polymer layers. UST increases this parameter; notably, the magnitude of AI changes is significantly larger in structures fabricated at lower spin-coating speeds, with longitudinal vibrations being more effective. For shunt resistance, the trends are nearly mirror symmetric, with the only distinction being a slightly stronger effect of radial vibrations. A comparison of Fig. 12a with Figs. 6b and 6d enables a clear correlation between J_{01}, R_p, and deep traps at 0.27–0.28 eV, which are likely associated with P_b centers. On the one hand, the presence of these levels enables carriers to overcome the energy barrier via trap-assisted tunneling (TAT), while on the other hand, their high concentration promotes parasitic shunt conduction pathways. In the D320 samples, these defects are absent, resulting in low J_{01} and high R_p. That is likely AI accelerated aging in thick polymer layers (Fig. 6b) significantly modifies both J_{01} and R_p. For thin layers, the influence of UST on both deep traps (Fig. 6d) and the J_{01} and R_p (Fig. 12a) is substantially weaker. The results indicate that the smoothing of spatial inhomogeneities induced by radial vibrations plays a more important role in R_p reducing than in governing TAT processes.

When discussing the impact of UST on J_{02} and n (Fig. 12b), several factors with potentially competing effects must be considered. In particular, changes in carrier lifetime associated with recombination play a key role in determining the recombination current density. For radial ultrasound, an almost one-to-one correlation is observed between J_{02} and τ_r obtained from impedance measurements. Specifically, for the D320uR sample, J_{02} increases by a factor of 1.6, while τ_r decreases by a factor of 1.3 relative to the pristine reference sample; for D520uR, the corresponding changes are factors of 5.2 and 3.6, respectively. For the uP mode, the reduction in recombination time is partially offset by an additional effect, which is particularly significant in samples with thin polymer layers. We attribute this effect to the formation of a PSS-rich interlayer, as previously discussed in the context of $R_{s,\mathrm{JV}}$.

As illustrated in Fig. 12b, the application of longitudinal UST leads to a marginal increase in the ideality factor, whereas radial UST induces a pronounced decrease in this parameter. On one hand, n is primarily governed by the defect density in the vicinity of the interface, which correlates well with admittance spectroscopy data (Fig. 7). In parallel, the ideality factor is known to scale with spatial inhomogeneity of the metal contact [85]. In the present case, improved contact quality is observed in structures incorporating a thin polymer layer, as evidenced by the reduced series resistance, extracted from IS measurements (Fig. 7). Notably, this mitigating effect is more pronounced under the radial UST. Consequently, the D520uP and D520uR series exhibit nearly identical ideality factors, despite the higher interface defect density observed in the latter.

It was suggested earlier that UST contributes to the acceleration of aging. To test this hypothesis, the additional measurements were performed on samples after 6 months of storage in the dark at room temperature. The changes in some parameters obtained from admittance spectroscopy (N_{total}), impedance spectroscopy (τ_r), and current–voltage measurements (R_p) are summarized in Table 1. As can be seen, irrespective of whether UST was applied, the values of these parameters after 6 months of aging are remarkably similar across all sample series. However, whereas in the control D320 series the total trap concentration changed by 94%, the characteristic recombination time decreased by 90 μs, and the shunt resistance decreased by nearly a factor of nine over this period, the corresponding changes in samples treated with longitudinal or radial acoustic waves were substantially smaller. The N_{total} changed by 43% and 28%, respectively, the τ_r changed by 40 μs, and R_p increased by factors of 1.5 and 2.3, respectively. Thus, ultrasound appears to drive the system toward a more stable state, characterized by substantially smaller temporal variations in the electrophysical parameters.

In summary, a comprehensive analysis of the Ag/PEDOT:PSS/n-Si/Zn-Ga structural parameters obtained using complementary techniques yields a coherent and physically consistent model of AI modifications governed by different ultrasound wave modes. The proposed mechanisms underlying the acoustically induced reconstruction of the polymer layers are inferred from electrophysical measurements and, in the absence of direct structural or chemical characterization, should therefore be regarded as working hypotheses rather than established facts. However, two considerations should be taken into account. On the one hand, direct structural characterization of the studied samples is subject to certain technical limitations. Since the experiments were performed on fully fabricated devices with metal contacts, direct examination of the underlying polymer morphology is prevented by the presence of the electrode, while its removal would inevitably disturb the intrinsic film structure. Moreover, soft polymer matrices are susceptible to beam-induced damage and ion-beam mixing during depth-profiling techniques such as XPS and ToF-SIMS. On the other hand, our proposed interpretations are well supported by previously reported direct experimental evidence. For example, Wang et al. [32] used AFM and a four-point probe method to demonstrate

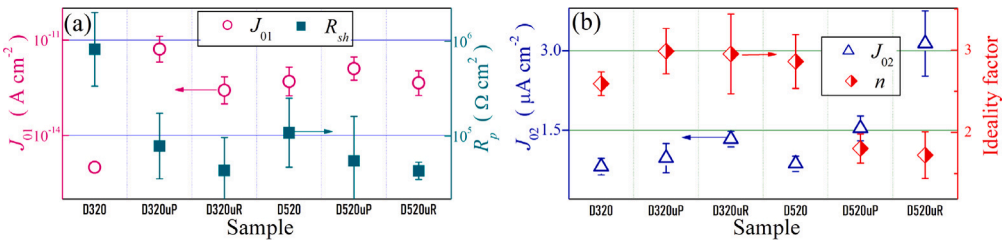


Fig. 12. Saturation current densities of the diffusion (a, open circles) and recombination (b, open triangles) current components, shunt resistance (a, filled squares), and ideality factor (b, semi-filled diamonds) for different sample series.

Table 1
Total trap concentrations, characteristic recombination times, and shunt resistances for samples with a thick polymer layer, determined immediately after fabrication and after 6 months of storage.

Sample	D320		D320uP		D320uR	
Sample state	As-prepared	Aged	As-prepared	Aged	As-prepared	Aged
N_{total} (10^{14} cm^{-3})	3.2 ± 0.9	6.2 ± 0.9	4.2 ± 0.9	6.0 ± 0.8	4.3 ± 0.9	5.5 ± 0.9
τ_r (μs)	170 ± 30	80 ± 20	60 ± 10	100 ± 10	130 ± 30	90 ± 10
R_p (10^4 Ω cm^2)	82 ± 6	9.1 ± 0.5	7.8 ± 0.5	12 ± 4	4.3 ± 0.3	9.8 ± 0.2

substrate-vibration-induced phase separation between PEDOT and PSS, accompanied by an increase in near-surface conductivity. Similarly, Zhu et al. [29] observed the smoothing of surface inhomogeneities in PEDOT films using scanning electron microscopy (SEM) and AFM, while Yazdi et al. [33] confirmed acoustically induced film homogenization through digital analysis of topographic SEM images. The convergence of the experimental datasets provides mutual validation and establishes a self-consistent framework that clarifies the underlying physical mechanisms.

4. Conclusion

A comprehensive analysis was carried out to evaluate the impact of ultrasound treatment applied during fabrication of n-Si/PEDOT:PSS heterostructures on their electrical properties. It was found that UST narrows the energy spectrum of near-surface defects by restructuring unstable trap configurations while simultaneously accelerating initial aging processes, which is important for ensuring long-term stability of the system parameters. In addition, ultrasound processing reduces both the relaxation time constants and the ideality factor, while increasing the recombination current density.

The application of longitudinal waves, which induce alternating stress perpendicular to the substrate, may disrupt Coulombic interactions and potentially trigger polymer phase separation. This segregation could result in a highly homogeneous PEDOT-rich interlayer at the interface (as suggested by an increased dispersion parameter), alongside a higher-resistivity bulk region potentially characterized by increased PSS content, which may subsequently contribute to the higher DC resistance. In contrast, radial oscillations, due to their in-plane components, may enhance macroscopic film homogeneity and potentially induce spatial anisotropy of PEDOT chains. Consequently, this improves metal contact quality, slows vertical carrier transport, and suppresses the contribution of parasitic capacitance associated with interface states. These effects scale inversely with polymer layer thickness.

The diversity of the observed phenomena highlights the feasibility of tailoring device performance through precise selection of the ultrasound processing regime. Thus, ultrasound represents an effective active tool for controlling the properties of organic–silicon heterostructures.

CRediT authorship contribution statement

Oleg Olikh: Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Denis Kalyuzhny:** Writing – original draft, Investigation, Formal analysis, Data curation. **Victor Kozachenko:** Validation, Resources, Methodology, Investigation. **Serhiy Kondratenko:** Writing – original draft, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.mssp.2026.111262.

Data availability

Data will be made available on request.

References

[1] J. Zhu, X. Yang, J. Sheng, P. Gao, J. Ye, Double-layered PEDOT:PSS films inducing strong inversion layers in organic/silicon hybrid heterojunction solar cells, ACS Appl. Energy Mater. 1 (6) (2018) 2874–2881, http://dx.doi.org/10.1021/acsaem.8b00533.

[2] L. Yu, R. Yang, X. Duan, Z. Wu, S. Zhang, X. Song, Y. Jiang, C. Xia, Molecular engineering of PEDOT:PSS for enhancing open-circuit voltage in organic/silicon heterojunction solar cells, Appl. Phys. Lett. 127 (26) (2025) 261601, http://dx.doi.org/10.1063/5.0306461.

[3] Z. Lu, J. Chen, Y. Wu, H. Wen, D. Chi, S. Huang, Dipole engineering in self-assembled monolayers for efficient organic–silicon hybrid solar cells, Appl. Phys. Lett. 127 (9) (2025) 093901, http://dx.doi.org/10.1063/5.0286438.

[4] J. Schmidt, V. Titova, D. Zielke, Organic-silicon heterojunction solar cells: Open-circuit voltage potential and stability, Appl. Phys. Lett. 103 (18) (2013) 183901, http://dx.doi.org/10.1063/1.4827303.

[5] Y. Da, X. Liu, Y. Xuan, Q. Li, Photon management effects of hybrid nanostructures/microstructures for organic-silicon heterojunction solar cells, Int. J. Energy Res. 42 (15) (2018) 4875–4890, http://dx.doi.org/10.1002/er.4249.

[6] D. Zielke, A. Pazidis, F. Werner, J. Schmidt, Organic-silicon heterojunction solar cells on n-type silicon wafers: The BackPEDOT concept, Sol. Energ. Mat. Sol. 131 (2014) 110–116, http://dx.doi.org/10.1016/j.solmat.2014.05.022, SI: SiliconPV 2014.

[7] S. Kondratenko, V. Lysenko, Y.V. Gomeniuk, O. Kondratenko, Y. Kozyrev, O. Selyshchev, V. Dzhagan, D.R.T. Zahn, Charge carrier transport, trapping, and recombination in PEDOT:PSS/n-Si solar cells, ACS Appl. Energy Mater. 2 (8) (2019) 5983–5991, http://dx.doi.org/10.1021/acsaem.9b01083.

[8] D. Zielke, R. Gogolin, M.-U. Halbich, C. Marquardt, W. Lövenich, R. Sauer, J. Schmidt, Large-area PEDOT:PSS/c-Si heterojunction solar cells with screen-printed metal contacts, Sol. RRL 2 (3) (2018) 1700191, http://dx.doi.org/10.1002/solr.201700191.

[9] Z. Yang, P. Gao, J. He, W. Chen, W.-Y. Yin, Y. Zeng, W. Guo, J. Ye, Y. Cui, Tuning of the contact properties for high-efficiency si/PEDOT:Pss heterojunction solar cells, ACS Energy Lett. 2 (3) (2017) 556–562, http://dx.doi.org/10.1021/acsenergylett.7b00015.

[10] J.P. Thomas, L. Zhao, M. Abd-Ellah, N.F. Heinig, K.T. Leung, Interfacial micropore defect formation in PEDOT:PSS-si hybrid solar cells probed by TOF-SIMS 3D chemical imaging, Anal. Chem. 85 (14) (2013) 6840–6845, http://dx.doi.org/10.1021/ac401084x.

[11] C. Zhang, Y. Zhang, H. Guo, Q. Jiang, P. Dong, C. Zhang, Efficient planar hybrid n-si/PEDOT:PSS solar cells with power conversion efficiency up to 13.31 interlayer, Energies 11 (6) (2018) http://dx.doi.org/10.3390/en11061397.

[12] X. Shen, Y. Zhu, T. Song, S.-T. Lee, B. Sun, Hole electrical transporting properties in organic-si schottky solar cell, Appl. Phys. Lett. 103 (1) (2013) 013504.

[13] J. Chen, L. Yang, K. Ge, B. Chen, Y. Shen, J. Guo, H. Liu, Y. Xu, J. Fan, Y. Mai, On the light-induced enhancement in photovoltaic performance of PEDOT:PSS/Si organic-inorganic hybrid solar cells, Appl. Phys. Lett. 111 (18) (2017) 183904, http://dx.doi.org/10.1063/1.5003993.

[14] S. Hu, Y. Yuan, C. Peng, L. Chen, X. Li, J. Zhang, Performance enhancement of solution-processed amorphous WZTO TFT with HAO gate dielectric via power ultrasound technology, Displays 78 (2023) 102408, http://dx.doi.org/10.1016/j.displa.2023.102408.

[15] J.C. Colmenares, E. Kuna, P. Lisowski, Synthesis of photoactive materials by sonication: Application in photocatalysis and solar cells, Top. Curr. Chem. 374 (5) (2016) 59, http://dx.doi.org/10.1007/s41061-016-0062-y.

[16] M. Heimann, P. Klaerner, C. Luechinger, A. Mette, J. Mueller, M. Traeger, T. Barthel, O. Valentin, P. Wawer, Ultrasonic bonding of aluminum ribbons to interconnect high-efficiency crystalline-silicon solar cells, Energy Procedia 27 (2012) 670–675, http://dx.doi.org/10.1016/j.egypro.2012.07.127, Proceedings of the 2nd International Conference on Crystalline Silicon Photovoltaics SiliconPV 2012.

[17] H. Basher, M.N. Zulkifli, A. Jalar, M. Daenen, Effect of ultrasonic bonding parameters on the contact resistance and bondability performances of CIGS thin film photovoltaic solar panel, IEEE J. Photovolt. 11 (2) (2021) 345–353, http://dx.doi.org/10.1109/JPHOTOV.2020.3047295.

[18] M. Brinkmann, T. Daschinger, R. Brendel, H. Schulte-Huxel, Ultrasonic tinning of al busbars for a silver-free rear side on bifacial silicon solar cells, IEEE J. Photovolt. 15 (2) (2025) 244–251, http://dx.doi.org/10.1109/JPHOTOV.2025.3533901.

[19] B. Romanjuk, V. Kladko, V. Melnik, V. Popov, V. Yukhymchuk, A. Gudymenko, Y. Olikh, G. Weidner, D. Krüger, Enhanced relaxation of SiGe layers by he implantation supported by in situ ultrasonic treatments, Mater. Sci. Semicond. Process. 8 (1–3) (2005) 171–175, http://dx.doi.org/10.1016/j.mssp.2004.09.030.

[20] V.G. Litovchenko, V.P. Melnik, B.M. Romanjuk, Nano-size phase formation at acoustically stimulated ion beam synthesis, Ukr. J. Phys. 60 (1) (2015) 64, http://dx.doi.org/10.15407/ujpe60.01.0064.

[21] S. Kalem, O. Yavuzcetin, C. Altineller, Effect of light exposure and ultrasound on the formation of porous silicon, J. Porous Mater. 7 (1) (2000) 381–383, http://dx.doi.org/10.1023/A:1009687021287.

[22] F. Soltani-kordshuli, F. Zabihi, M. Eslamian, Graphene-doped PEDOT:PSS nanocomposite thin films fabricated by conventional and substrate vibration-assisted spray coating (SVASC), Eng. Sci. Technol. an Int. J. 19 (3) (2016) 1216–1223, http://dx.doi.org/10.1016/j.jestch.2016.02.003.

[23] K. Anand, R. Kaur, S. Tripathi, Tuning of photoinduced resistive switching characteristics of MoS2/PVA photomemristor via ultrasonication, J. Alloys Compd. 1020 (2025) 179307, http://dx.doi.org/10.1016/j.jallcom.2025.179307.

[24] N. Gholampour, M. Eslamian, Ultrasound-assisted synthesis of layered zeolitic imidazolate framework: crystal formation and characteristics, J. Coord. Chem. 73 (2) (2020) 317–332, http://dx.doi.org/10.1080/00958972.2020.1713316.

[25] S. Waheed, S. Pareek, P. Singh, P. Sharma, A. Rana, S. Karak, Effect of in situ annealing on phase segregation and optoelectronic properties of ultrasonic-spray deposited polymer blend films, IEEE J. Photovolt. 10 (6) (2020) 1727–1734, http://dx.doi.org/10.1109/JPHOTOV.2020.2983304.

[26] J. Zhao, X. Yang, W. Zhou, R. Wang, Y. Wang, J. Zhang, X. Zhong, H. Zhang, H. Ren, Y. Zhao, X. Zhang, G. Hou, B. Li, Y. Ding, Chloride additive optimization in ultrasonic spray coating process for high-performance perovskite solar cell realization, Prog. Photovoltaics Res. Appl. 34 (4) (2026) 429–438, http://dx.doi.org/10.1002/pip.70050.

[27] J.-J. Huang, Y.-R. Ho, Y.-Q. Wu, Effect of substrate temperature on the performance of ultrasonically sprayed PEDOT:PSS–agnws transparent conductive films for electrochromic devices, J. Mater. Sci.: Mater. Electron. 33 (24) (2022) 19490–19500, http://dx.doi.org/10.1007/s10854-022-08784-8.

[28] S. Waheed, S. Pareek, P. Sharma, S. Karak, Performance improvement of ultrasonic spray deposited polymer solar cell through droplet boundary reduction assisted by acoustic substrate vibration, Semicond. Sci. Technol. 36 (1) (2020) 015002, http://dx.doi.org/10.1088/1361-6641/abbba8.

[29] Y. Zhu, R. Ravishekar, T. Tang, B. Gogoi, C. Gockley, S. Venu, T.L. Alford, X. Li, Characterization of PEDOT:PSS nanofilms printed via electrically assisted direct ink deposition with ultrasonic vibrations, Molecules 28 (20) (2023) http://dx.doi.org/10.3390/molecules28207109.

[30] F. Zabihi, M.-R. Ahmadian-Yazdi, M. Eslamian, Photocatalytic graphene-TiO2 thin films fabricated by low-temperature ultrasonic vibration-assisted spin and spray coating in a sol-gel process, Catalysts 7 (5) (2017) http://dx.doi.org/10.3390/catal7050136.

[31] F. Zabihi, M.-R. Ahmadian-Yazdi, M. Eslamian, Fundamental study on the fabrication of inverted planar perovskite solar cells using two-step sequential substrate vibration-assisted spray coating (2S-SVASC), Nanoscale Res. Lett. 11 (1) (2016) 71, http://dx.doi.org/10.1186/s11671-016-1259-2.

[32] Q. Wang, M. Eslamian, Improving uniformity and nanostructure of solution-processed thin films using ultrasonic substrate vibration post treatment (SVPT), Ultrasonics 67 (2016) 55–64, http://dx.doi.org/10.1016/j.ultras.2015.12.012.

[33] M.-R. Ahmadian-Yazdi, C. Barratt, A. Rahimzadeh, M. Eslamian, Microstructural and nanostructural evolution of light harvester perovskite thin film under the influence of ultrasonic vibrations, ACS Omega 5 (1) (2020) 808–821, http://dx.doi.org/10.1021/acsomega.9b03566.

[34] Y. Wang, M.-R. Ahmadian-Yazdi, Y. Ni, Y. Jiang, M. Eslamian, Z. Jin, Q. Chen, Boosted charge-carrier transport in triple-cation perovskites by ultrasonic vibration post treatment, Adv. Electron. Mater. 8 (7) (2022) 2101286, http://dx.doi.org/10.1002/aelm.202101286.

[35] X. Zhang, F. Zabihi, H. Xiong, M. Eslamian, C. Hou, M. Zhu, H. Wang, Q. Zhang, Highly efficient flexible perovskite solar cells made via ultrasonic vibration assisted room temperature cold sintering, Chem. Eng. J. 394 (2020) 124887, http://dx.doi.org/10.1016/j.cej.2020.124887.

[36] N. Gholampour, D. Brian, M. Eslamian, Tailoring characteristics of PEDOT:PSS coated on glass and plastics by ultrasonic substrate vibration post treatment, Coatings 8 (10) (2018) http://dx.doi.org/10.3390/coatings8100337.

[37] Y. Xie, F. Zabihi, M. Eslamian, Fabrication of highly reproducible polymer solar cells using ultrasonic substrate vibration posttreatment, J. Photonics Energy 6 (4) (2016) 045502, http://dx.doi.org/10.1117/1.JPE.6.045502.

[38] S. Waheed, S. Pareek, T. Abhijith, R. Suthar, P. Sharma, S. Karak, Understanding the influences of in-situ annealing and substrate vibration on the charge carrier dynamics of ultrasonic spray-coated polymer solar cell, J. Mater. Sci.: Mater. Electron. 33 (18) (2022) 15180–15190, http://dx.doi.org/10.1007/s10854-022-08437-w.

[39] X. Qu, J. Ma, K. Wang, X.W. Sun, Characteristic voltages and times from capacitance–voltage analysis of quantum dot light-emitting diodes, Appl. Phys. Lett. 124 (26) (2024) 263503, http://dx.doi.org/10.1063/5.0221019.

[40] X. Qu, X. Sun, Impedance spectroscopy for quantum dot light-emitting diodes, J. Semicond. 44 (9) (2023) 091603, http://dx.doi.org/10.1088/1674-4926/44/9/091603.

[41] L. Zhang, H. Nakanotani, C. Adachi, Capacitance-voltage characteristics of a 4,4-bis[(N-carbazole)styryl]biphenyl based organic light-emitting diode: Implications for characteristic times and their distribution, Appl. Phys. Lett. 103 (9) (2013) 093301, http://dx.doi.org/10.1063/1.4819436.

[42] X. Fan, N.E. Stott, J. Zeng, Y. Li, J. Ouyang, L. Chu, W. Song, PEDOT:PSS materials for optoelectronics, thermoelectrics, and flexible and stretchable electronics, J. Mater. Chem. A 11 (2023) 18561–18591, http://dx.doi.org/10.1039/D3TA03213B.

[43] S. Jäckle, M. Mattiza, M. Liebhaber, G. Brönstrup, M. Rommel, K. Lips, S. Christiansen, Junction formation and current transport mechanisms in hybrid n-si/PEDOT:PSS solar cells, Sci. Rep. 5 (1) (2015) 13008, http://dx.doi.org/10.1038/srep13008.

[44] F. Fabregat-Santiago, G. Garcia-Belmonte, I. Mora-Seró, J. Bisquert, Characterization of nanostructured hybrid and organic solar cells by impedance spectroscopy, Phys. Chem. Chem. Phys. 13 (2011) 9083–9118, http://dx.doi.org/10.1039/C0CP02249G.

[45] T. Walter, R. Herberholz, C. Müller, H.W. Schock, Determination of defect distributions from admittance measurements and application to cu(in,ga)se2 based heterojunctions, J. Appl. Phys. 80 (8) (1996) 4411–4420, http://dx.doi.org/10.1063/1.363401.

[46] M. Zhang, F. Guo, S. Lei, T. Zhong, B. Xiao, C. Liu, L. Wang, J. Chen, Q. You, J. Liu, R. Yang, Positive temperature dependence of the electroluminescent performance in a colloidal quantum dot light-emitting diode, Dye. Pigm. 195 (2021) 109703, http://dx.doi.org/10.1016/j.dyepig.2021.109703.

[47] J.V. Li, G. Ferrari (Eds.), Capacitance Spectroscopy of Semiconductors, first ed., Jenny Stanford Publishing, 2018, http://dx.doi.org/10.1201/b22451.

[48] J.T. Heath, P. Zabierowski, Capacitance spectroscopy of thin-film solar cells, in: D. Abou-Ras, T. Kirchartz, U. Rau (Eds.), Advanced Characterization Techniques for Thin Film Solar Cells, Wiley-VCH, 2011, pp. 81–105, http://dx.doi.org/10.1002/9783527636280.ch4.

[49] Y. Dou, S. Luo, P. Zhu, L. Zhu, G. Zhang, C. Bao, H. Yan, J. Zhu, S. Chen, Distributions and evolution of trap states in non-fullerene organic solar cells, Joule 9 (1) (2025) http://dx.doi.org/10.1016/j.joule.2024.10.006.

[50] F. Ye, Z.-C. Zhao, C.-S. He, J.-L. Liang, Q. Gao, Y.-Z. Xie, D.-P. Zhang, X.-M. Cai, Capacitance characterization and current transport mechanism of Znsnn2 heterojunctions, Appl. Phys. Lett. 125 (24) (2024) 242102, http://dx.doi.org/10.1063/5.0241401.

[51] J.A. Carr, M. Elshobaki, S. Chaudhary, Deep defects and the attempt to escape frequency in organic photovoltaic materials, Appl. Phys. Lett. 107 (20) (2015) 203302, http://dx.doi.org/10.1063/1.4936160.

[52] D. Bozyigit, S. Volk, O. Yarema, V. Wood, Quantification of deep traps in nanocrystal solids, their electronic properties, and their influence on device behavior, Nano Lett. 13 (11) (2013) 5284–5288, http://dx.doi.org/10.1021/nl402803h.

[53] N.S. Ndiaye, O. Simonetti, T.-P. Nguyen, L. Giraudet, Generation-recombination in disordered organic semiconductor: Application to the characterization of traps, Org. Electron. 99 (2021) 106350, http://dx.doi.org/10.1016/j.orgel.2021.106350.

[54] Z. Wang, H. Wu, H. Li, J. Yang, F.C. Ling, L. Wang, A practical guide to electrical characterization of interface states: Case studies on SiC and GaN, Mater. Sci. Semicond. Process. 201 (2026) 110050, http://dx.doi.org/10.1016/j.mssp.2025.110050.

[55] Q. Chen, J.E. Sanchez, D. Lin, Y. Lei, G. Zhu, Analytical model for donor like Gaussian traps in organic thin-film transistor, Org. Electron. 103 (2022) 106464, http://dx.doi.org/10.1016/j.orgel.2022.106464.

[56] P. Liu, Y. Huang, Z. Wang, W. Liu, B. Yap, Z. He, H. Wu, Evolution of interfacial defects and energy losses during aging of organic photovoltaics, Phys. B Condens. Matter 677 (2024) 415707, http://dx.doi.org/10.1016/j.physb.2024.415707.

[57] M. Zhang, F. Guo, Q. Zhou, T. Zhong, B. Xiao, L. Zou, Q. You, B. You, Y. Li, X. Liu, H. Liu, J. Yan, J. Liu, Enhanced performance through trap states passivation in quantum dot light emitting diode, J. Lumin. 234 (2021) 117946, http://dx.doi.org/10.1016/j.jlumin.2021.117946.

[58] L. Xu, J. Wang, J.W.P. Hsu, Transport effects on capacitance-frequency analysis for defect characterization in organic photovoltaic devices, Phys. Rev. Appl. 6 (2016) 064020, http://dx.doi.org/10.1103/PhysRevApplied.6.064020.

[59] J. Li, G. Liu, W. Liu, Y. Si, W. Deng, H. Wu, Highly sensitive UV–Vis-to-near-infrared organic photodetectors employing ZnO: Polyethylenimine ethoxylated composite as hole-blocking layer, Adv. Photonics Res. 3 (4) (2022) 2100269, http://dx.doi.org/10.1002/adpr.202100269.

[60] T. Zhang, Y. Xu, H. Yao, J. Zhang, P. Bi, Z. Chen, J. Wang, Y. Cui, L. Ma, K. Xian, Z. Li, X. Hao, Z. Wei, J. Hou, Suppressing the energetic disorder of all-polymer solar cells enables over 18% efficiency, Energy Env. Sci. 16 (4) (2023) 1581–1589, http://dx.doi.org/10.1039/d2ee03535a.

[61] W. Zuo, S. Liu, H. Li, L. Zhang, W. Wu, J. Chen, J. Peng, Capacitance–voltage extraction method for the deep-level defect distribution in organic photodiode, Appl. Phys. Lett. 123 (26) (2023) 263504, http://dx.doi.org/10.1063/5.0179058.

[62] H. Yoo, M. Ghittorelli, D.-K. Lee, E.C.P. Smits, G.H. Gelinck, H. Ahn, H.-K. Lee, F. Torricelli, J.-J. Kim, Balancing hole and electron conduction in ambipolar split-gate thin-film transistors, Sci. Rep. 7 (1) (2017) 5015, http://dx.doi.org/10.1038/s41598-017-04933-w.

[63] S.V. Yampolskii, Y.A. Genenko, C. Melzer, H. von Seggern, Self-consistent model of unipolar transport in organic semiconductor diodes: Accounting for a realistic density-of-states distribution, J. Appl. Phys. 109 (7) (2011) 073722, http://dx.doi.org/10.1063/1.3569839.

[64] Q. Niu, H. Zhang, K. Li, An improved TLBO with elite strategy for parameters identification of PEM fuel cell and solar cell models, Int. J. Hydrog. Energy 39 (8) (2014) 3837–3854, http://dx.doi.org/10.1016/j.ijhydene.2013.12.110.

[65] O. Olikh, A test of meta-heuristic algorithms for parameter extraction of next-generation solar cells with S-shaped current–voltage curves, Mater. Sci. Eng.: B 307 (2024) 117506, http://dx.doi.org/10.1016/j.mseb.2024.117506.

[66] M. Mikelsen, J.H. Bleka, J.S. Christensen, E.V. Monakhov, B.G. Svensson, J. Härkönen, B.S. Avset, Annealing of the divacancy-oxygen and vacancy-oxygen complexes in silicon, Phys. Rev. B 75 (2007) 155202, http://dx.doi.org/10.1103/PhysRevB.75.155202.

[67] A. Chroneos, C.A. Londos, E.N. Sgourou, P. Pochet, Point defect engineering strategies to suppress A-center formation in silicon, Appl. Phys. Lett. 99 (24) (2011) 241901, http://dx.doi.org/10.1063/1.3666226.

[68] R. Wang, Y. Wang, C. Wu, T. Zhai, J. Yang, B. Sun, S. Duhm, N. Koch, Direct observation of conductive polymer induced inversion layer in n-si and correlation to solar cell performance, Adv. Funct. Mater. 30 (4) (2020) 1903440, http://dx.doi.org/10.1002/adfm.201903440.

[69] S. Jäckle, M. Liebhaber, C. Gersmann, M. Mews, K. Jäger, S. Christiansen, K. Lips, Potential of PEDOT:PSS as a hole selective front contact for silicon heterojunction solar cells, Sci. Rep. 7 (1) (2017) 2170, http://dx.doi.org/10.1038/s41598-017-01946-3.

[70] H. Zhu, W. Xie, X. Liu, Y. Liu, J. Zhang, X. Zuo, First-principles study of non-radiative carrier capture by defects at amorphous-SiO2/Si(100) interface, Chin. Phys. B 32 (7) (2023) 077303, http://dx.doi.org/10.1088/1674-1056/ac9fc2.

[71] I. Capan, V. Janicki, R. Jacimovic, B. Pivac, C–V and DLTS studies of radiation induced si–SiO2 interface defects, Nucl. Instrum. Methods Phys. Res. Sect. B Beam Interact. Mater. At. 282 (2012) 59–62, http://dx.doi.org/10.1016/j.nimb.2011.08.065, Ion Beam Synthesis and Modification of Nanostructured Materials and Surfaces.

[72] J. Chen, Y. Shen, J. Guo, B. Chen, J. Fan, F. Li, B. Liu, H. Liu, Y. Xu, Y. Mai, Electrochemical grafting passivation of silicon via electron transfer at polymer/silicon hybrid interface, Electrochim. Acta 247 (2017) 826–834, http://dx.doi.org/10.1016/j.electacta.2017.07.071.

[73] P. Li, Z. Chen, P. Yao, F. Zhang, J. Wang, Y. Song, X. Zuo, First-principles study of defects in amorphous-SiO2/Si interfaces, Appl. Surf. Sci. 483 (2019) 231–240, http://dx.doi.org/10.1016/j.apsusc.2019.03.216.

[74] A. Stirling, A. Pasquarello, J.-C. Charlier, R. Car, Dangling bond defects at $Si-SiO_2$ interfaces: Atomic structure of the P_b1 center, Phys. Rev. Lett. 85 (2000) 2773–2776, http://dx.doi.org/10.1103/PhysRevLett.85.2773.

[75] E. von Hauff, D. Klotz, Impedance spectroscopy for perovskite solar cells: characterisation, analysis, and diagnosis, J. Mater. Chem. C 10 (2) (2022) 742–761, http://dx.doi.org/10.1039/d1tc04727b.

[76] B.J. Leever, C.A. Bailey, T.J. Marks, M.C. Hersam, M.F. Durstock, In situ characterization of lifetime and morphology in operating bulk heterojunction organic photovoltaic devices by impedance spectroscopy, Adv. Energy Mater. 2 (1) (2012) 120–128, http://dx.doi.org/10.1002/aenm.201100357.

[77] D. Cotfas, P. Cotfas, S. Kaplanis, Methods and techniques to determine the dynamic parameters of solar cells: Review, Renew. Sustain. Energy Rev. 61 (2016) 213–221, http://dx.doi.org/10.1016/j.rser.2016.03.051.

[78] E. von Hauff, Impedance spectroscopy for emerging photovoltaics, J. Phys. Chem. C 123 (18) (2019) 11329–11346, http://dx.doi.org/10.1021/acs.jpcc.9b00892.

[79] O. Oklobia, S. Komilian, T. Sadat-Shafai, Impedance spectroscopy and capacitance – voltage measurements analysis: Impact of charge carrier lifetimes and mapping vertical segregation in bulk heterojunction P3HT: PCBM solar cells, Org. Electron. 61 (2018) 276–281, http://dx.doi.org/10.1016/j.orgel.2018.06.003.

[80] Y. Yoon, H.J. Kim, C.-H. Cho, S. Kim, H.J. Son, M.-J. Ko, H. Kim, D.-K. Lee, J.Y. Kim, W. Lee, B.J. Kim, B. Kim, Carrier lifetime extension via the incorporation of robust hole/electron blocking layers in bulk heterojunction polymer solar cells, ACS Appl Mater Interfaces 6 (1) (2014) 333–339, http://dx.doi.org/10.1021/am404381e.

[81] B. Hailegnaw, N.S. Sariciftci, M.C. Scharber, Impedance spectroscopy of perovskite solar cells: Studying the dynamics of charge carriers before and after continuous operation, Phys. Status Solidi (A) 217 (22) (2020) 2000291, http://dx.doi.org/10.1002/pssa.202000291.

[82] A.M.C. Gomes, D.R. Huanca, Optical and electrical characterization of Er/Yb-doped porous silicon for advanced photovoltaic applications, Phys. Status Solidi (A) 222 (22) (2025) 2500414, http://dx.doi.org/10.1002/pssa.202500414.

[83] G. Friesen, M. Özsar, E. Dunlop, Impedance model for CdTe solar cells exhibiting constant phase element behaviour, Thin Solid Films 361–362 (2000) 303–308, http://dx.doi.org/10.1016/S0040-6090(99)00764-6.

[84] A.J. Bühler, A. Krenzinger, Method for photovoltaic parameter extraction according to a modified double–diode model, Prog. Photovolt., Res. Appl. 21 (5) (2013) 884–893, http://dx.doi.org/10.1002/pip.2170.

[85] A.S.H. van der Heide, A. Schonecker, J.H. Bultman, W.C. Sinke, Explanation of high solar cell diode factors by nonuniform contact resistance, Prog. Photovolt., Res. Appl. 13 (1) (2005) 3–16, http://dx.doi.org/10.1002/pip.556.